

AI Architecture and Design

Course Project

Human-Centered Evaluation and Redesign of an Existing AI System

1 Project Overview

Artificial Intelligence systems are increasingly used to support human decision-making, communication, education, healthcare, productivity, recommendation, and many other activities. However, a technically capable AI model does not automatically result in a usable AI system. The design of the interaction between the human and the AI is equally important.

In this project, you will select an existing AI-powered application and systematically evaluate it from a **Human-Centered Artificial Intelligence (HCAI)** and **Human-Computer Interaction (HCI)** perspective.

The primary purpose of this project is **not simply to evaluate the accuracy of the underlying AI model**. Instead, you will evaluate how effectively the complete AI system supports its users.

The central question for the project is:

How well does the AI system support users in understanding, interacting with, evaluating, and acting on AI-generated outputs?

You will identify strengths and weaknesses in the existing design and propose human-centered improvements to the system.

2 Learning Objectives

After completing this project, you should be able to:

- Apply HCI principles to an existing AI application.
- Identify the users, stakeholders, tasks, and context of use of an AI system.
- Analyze interaction between humans and AI systems.
- Evaluate an AI application using established usability and design principles.
- Analyze the division of responsibility between the human and the AI.
- Identify design problems using evidence rather than personal preference.
- Develop human-centered recommendations for improving an AI system.
- Communicate evaluation findings through a professional report, prototype, and presentation.

3 Part I: Select an Existing AI Application

Select an existing AI-powered application that you can interact with directly.

Possible applications include:

- Generative AI assistants, AI writing assistants, AI image-generation applications, AI coding assistants, AI recommendation systems, AI educational or tutoring applications, AI translation systems, AI career or resume applications, AI customer-service chatbots, AI productivity applications and AI-powered design applications

The selected application must provide sufficient interaction between a **human user and an AI system**.

Important: The project should evaluate the complete AI application and its interaction with users rather than simply describing the AI algorithm or machine-learning model used by the application.

4 Part II: AI System Description

Provide an overview of the selected AI application.

Your discussion should include:

1. Name of the application.
2. Purpose of the application.
3. Problem the application is designed to solve.
4. Major AI capabilities.
5. Inputs provided by users.
6. Outputs generated by the AI.
7. How users interact with the system.
8. Major decisions or actions supported by the system.

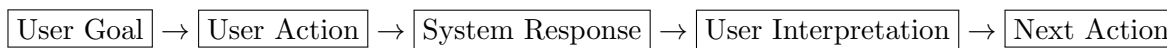
5 Part III: Users and Stakeholders

Identify the major users and stakeholders associated with the AI system.

6 Part IV: User Goals and Task Analysis

Identify at least **three important tasks** that users perform with the selected AI application.

For each task, describe the complete interaction:



For each task:

1. State the user's goal.
2. Describe the input provided by the user.
3. Describe the AI's response.
4. Explain how the user is expected to interpret the response.
5. Identify the next action available to the user.
6. Identify any difficulties or points of confusion.

7 Part V: Interaction Design Evaluation

Evaluate how effectively the interface communicates the interaction state to the user.

At different stages of interaction, determine whether the user can answer the following four questions:

1. **Where am I?**
2. **What can I do?**
3. **What will happen next?**
4. **What have I already done?**

For each question, provide specific evidence from the application.

Screenshots may be included where appropriate and permitted.

Do not simply state that the interface is “good” or “bad.” Explain **why** a particular design supports or interferes with user interaction.

8 Part VI: Evaluation Using HCI Design Rules

Evaluate the AI application using the following HCI design principles.

Principle	Evaluation Question
Learnability	Can a new user understand how to begin using the AI system and accomplish important tasks?
Predictability	Can users predict what will happen after performing an action?

Principle	Evaluation Question
Synthesizability	Can users understand the effects of their previous actions?
Familiarity	Does the system use concepts, terminology, and interactions that users already understand?
Consistency	Are similar actions, controls, and AI outputs presented consistently?
Observability	Can users determine the current state of the system and understand what the AI is doing?
Recoverability	Can users recover from mistakes, incorrect inputs, or unwanted AI outputs?
Responsiveness	Does the system provide timely and meaningful responses to user actions?
Task Conformance	Does the system support the tasks users actually need to accomplish?
Customizability	Can the interaction or interface be adapted appropriately to user needs?

For each principle, provide:

- A rating from **1–5**
- An explanation of the observed design
- A recommendation where improvement is needed

Use the following rating scale:

Rating	Meaning
1	Very Poor
2	Poor
3	Acceptable
4	Good
5	Excellent

9 Part VII: Feedback and System Status

Evaluate how effectively the AI system provides feedback.

Consider the following:

- Does the system acknowledge user input?
- Does the system indicate when AI processing is occurring?
- Does the system indicate when processing is complete?
- Does the system communicate errors clearly?
- Does the system explain why an action failed?

- Does the system indicate what the user can do next?

Identify at least:

- One example of effective feedback.
- One example of feedback that could be improved.

10 Part VIII: Cognitive Principles

Evaluate the system using at least **three** of the following cognitive principles discussed in class:

Attention Does the interface direct the user's attention toward important information?

Working Memory

Does the interface require users to remember too much information?

Recognition vs. Recall

Is important information visible, or must users remember it?

Visual Perception

Is information easy to perceive and distinguish?

Visual Hierarchy

Are the most important AI outputs visually prominent?

Mental Models

Can users develop an accurate understanding of how the AI system behaves?

Decision Making

Does the presentation of AI information support effective human decisions?

For each selected principle, provide evidence from the application and explain its impact on the user.

11 Part IX: Accessibility Evaluation

Evaluate the accessibility of the AI application.

Consider:

- Color contrast
- Text readability
- Font size
- Keyboard navigation

- Screen-reader compatibility where testable
- Captions or transcripts
- Alternative text
- Reliance on color alone
- Clear labels
- Cognitive accessibility

Provide at least **three accessibility observations** and recommendations where appropriate.

12 Part X: Human-Centered Redesign

Select **2–3 important problems** identified during your evaluation and state how you will incorporate new changes into your design.

13 Final Deliverables

The project consists of the following deliverables.

1. Project Proposal

This should be a one-page report summarizing the AI application system you have selected for your project. Your report should briefly describe the name of the AI application, its purpose, its primary users, and its main AI capabilities. Please also include a screenshot of the AI application system/interface in this report.

2. Project Report

Submit a professional report of approximately **6–10 pages**, excluding references and appendices.

The report should include:

- AI application description
- Users and stakeholders
- User goals and task analysis
- Interaction design evaluation
- HCI design-rules evaluation
- Human-AI interaction analysis

- Cognitive-principles evaluation
- Accessibility
- Prioritized design problems
- Redesign recommendations
- Conclusion

3. Presentation

Prepare an approximately **10–12 minute presentation**.

The presentation should communicate:

- The selected AI application
- Primary users and tasks
- Major HCI findings
- Evidence supporting the findings
- Human-AI interaction issues
- Usability results
- Major risks
- Proposed redesign
- Key lessons learned

14 Grading Rubric

Component	Weight
AI System, Users, and Task Analysis	2
HCI and Design-Rules Evaluation	4
Human-AI Interaction/ Feedback and System status	3
Accessibility	3
Cognitive principles	4
Redesign and Recommendations	2
Proposal	2
Professional Quality of Report and Presentation	5
Total	25

15 Important Expectations

This project is an **evaluation and analysis project**, not simply a description of an AI application. Strong projects will:

- Support conclusions with specific evidence.
- Apply HCI concepts discussed in class.
- Distinguish AI model capability from interface and interaction quality.
- Consider both strengths and weaknesses of the application.
- Consider the complete human-AI interaction process.
- Identify meaningful risks rather than superficial interface problems.
- Prioritize problems based on their impact on users.
- Provide realistic and well-justified redesign recommendations.

A statement such as:

“The interface is easy to use.”

is not sufficient by itself.

Instead, provide evidence and connect the observation to an HCI principle.

For example:

“The interface demonstrates good observability because it immediately indicates when the AI begins processing the user’s request and provides visible feedback when generation is complete.”